HANDS-ON 7 · DAY 2 / AFTERNOON · GROUP'S SHEET

Does any model beat the last value?

The case. The treasury team of Hands-on 6 buys an index fund once a month, on a day of its choosing, and now asks for next week's closes of the Dow Jones. Dataiku offers baselines, which fit nothing, and models, which do. A model earns its place only if it forecasts the closes better than the last close, on the same days.

DATA CARD · DOWJONES102425.CSV

What a row is. One trading day of the Dow Jones Industrial Average: 256 rows, no missing value. Used here: `Date`, read as a date in Hands-on 6 (step 1.2), and `Close`, the index at the day's end, in points. `Open`, `High`, `Low` and `Volume` are left aside.

The series forecast. `Close`, seven days ahead. No other column enters: a model fed `Volume` would need next week's volume too.

How the rows were chosen. Every trading day from 1 October 2024 to 8 October 2025: 373 calendar days, of which 117 have no close (the weekends and 11 market holidays).

Its history. Downloaded for the previous edition of the module and used unchanged. A one-day time step adds the 117 days without a close; step 1.3 fills each with the previous close.

Source and licence. stooq.com, symbol `^DJI`, daily. No licence is recorded: used in class, never republished.

From panel 4. `bike-rentals.csv`, uploaded in Hands-on 6: Capital Bikeshare's daily rentals in Washington DC, 2011 and 2012; `Casual` counts the riders who were not registered. H. Fanaee-T (2013), UCI dataset 275, CC BY 4.0.

file	rows	period	series forecast	used in
`dowjones102425.csv`	256 trading days	10.2024 to 10.2025	`Close`, by `Date`	steps 1 to 3
`bike-rentals.csv`	731 days	2011 and 2012	`Casual`, by `Date`	panel 4

STEP 1 · SET UP THE FORECAST

1 ☐ **Open a forecasting analysis**
On the Dow Jones dataset uploaded in Hands-on 6: **Lab** → **Time Series Forecasting**, target `Close`, time variable `Date`, a single time series; **Create**.

2 ☐ **Set the step and the horizon**
Design → **General settings**: **Time step** `1` **Day**; **Forecast horizon** `7`; **Skipped time steps in evaluation** `0`.

3 ☐ **Fill the missing days, split in time**
Design → **Train / Test Set**: **Numerical interpolation** set to the previous value, never **Linear**, which would borrow the next trading day's close; **K-fold cross-test** ticked, **Number of folds** `10`, **Fold offset** ticked.

4 ☐ **Score with MASE**
Design → **Metrics**: **Mean Absolute Scaled Error**.

STEP 2 · TRAIN THE BASELINES FIRST

1 ☐ **Choose the baselines**
Design → **Algorithms**: **Trivial identity** and **Seasonal naive**, season length `7`, and nothing else; **Train**.

Trivial identity forecasts every day ahead with: _____

STEP 3 · THEN THE MODELS

1 ☐ **Add the models**
Design → **Algorithms**: **AutoARIMA**, season length 7, and **ETS**, the baselines unticked; **Train**.

2 ☐ **Set each beside Trivial identity**
Results: each model's MASE and its ±, on the same folds.

forecast	MASE	±	below Trivial identity by more than the ±?
Trivial identity			(the benchmark)
Seasonal naive			
AutoARIMA			
ETS			

3 ☐ **Read what AutoARIMA chose**
On the AutoARIMA model: **Model information** → **ARIMA orders**. As on slides 11 to 16 of **Forecasting a time series**: **d** counts the differences, **p** the past values and **q** the past forecast errors the model uses; **P**, **D** and **Q** are the same three for the weekly season.

p, d, q: _____ P, D, Q: _____

With these orders, what does AutoARIMA forecast for tomorrow's close, and which baseline of step 2 forecasts the same?

_____

4 · RUN THE SAME SESSION ON THE CASUAL RIDERS

1 ☐ **Open the riders' forecast**
On the bike rentals dataset uploaded in Hands-on 6: **Lab** → **Time Series Forecasting**, target `Casual`, time variable `Date`; the settings of steps 1.2 to 1.4.

2 ☐ **Train the baselines and ETS**
Design → **Algorithms**: **Trivial identity**, **Seasonal naive** (season length 7) and **ETS**; **Train**.

`Casual`	MASE	±	below Trivial identity by more than the ±?
Trivial identity			(the benchmark)
Seasonal naive			
ETS			

5 · DECIDE

What do you tell the treasury team about next week's closes? Where did a model beat the last value, and what does that series have that the index lacks?